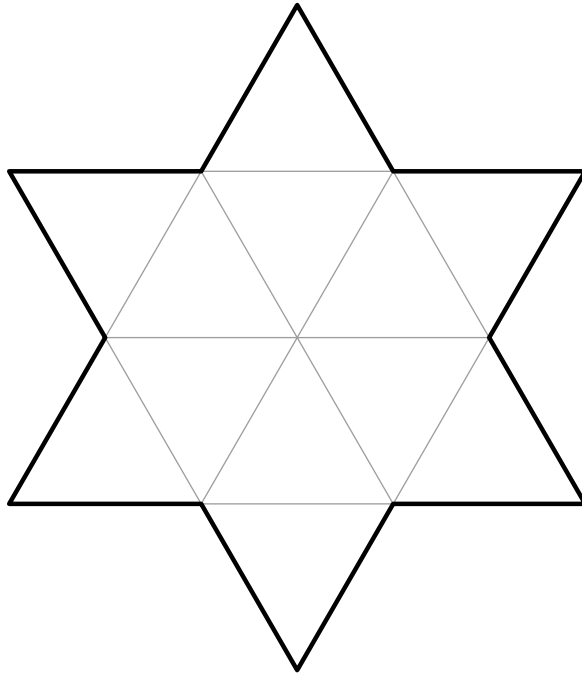
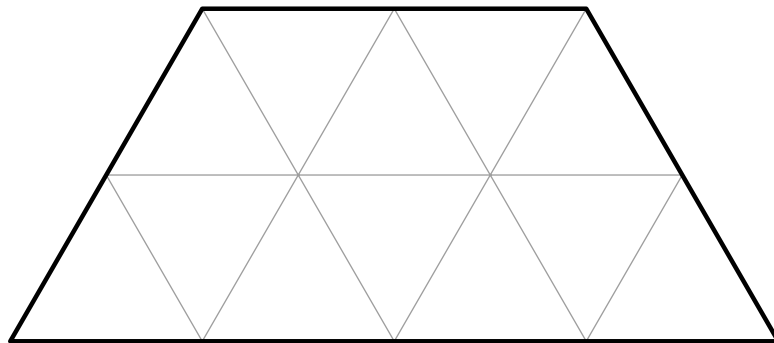
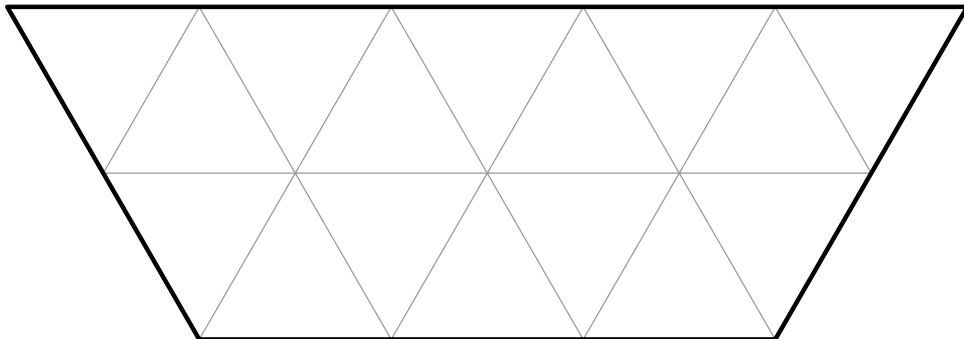
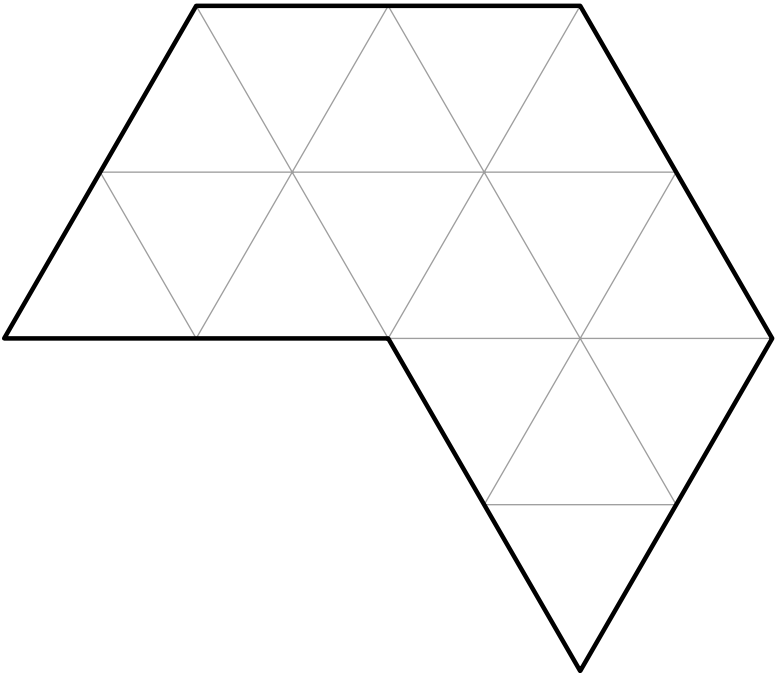


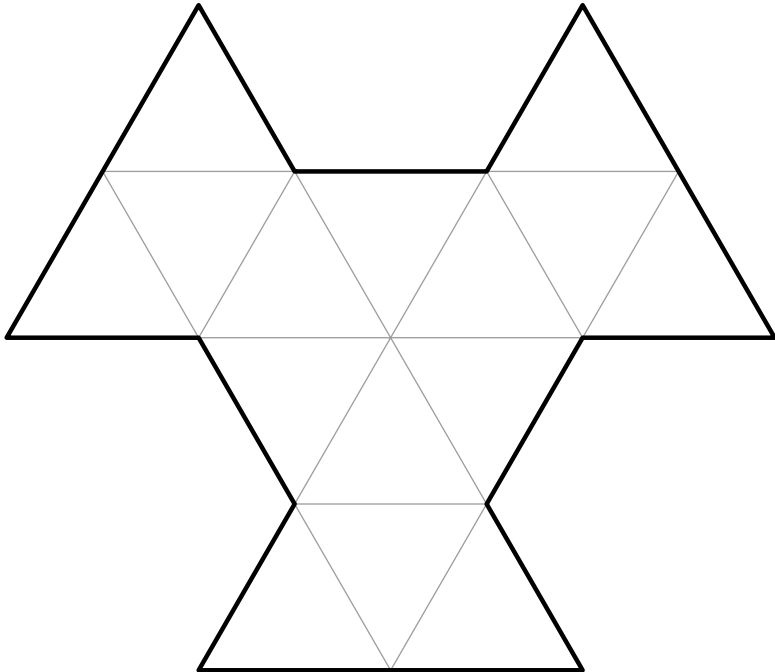
Problem 1: A board is covered when every small triangle on it is under a piece. Pieces may not overlap or stick out past the edge of the board. Which of boards A to E can be covered with blue pieces only? For a board that cannot be covered, how can you be sure?

A**B****C**

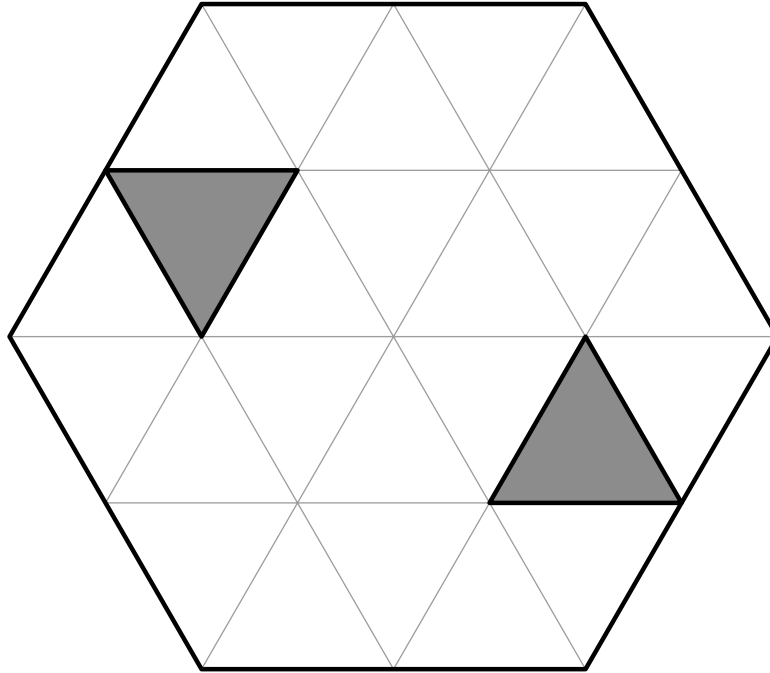
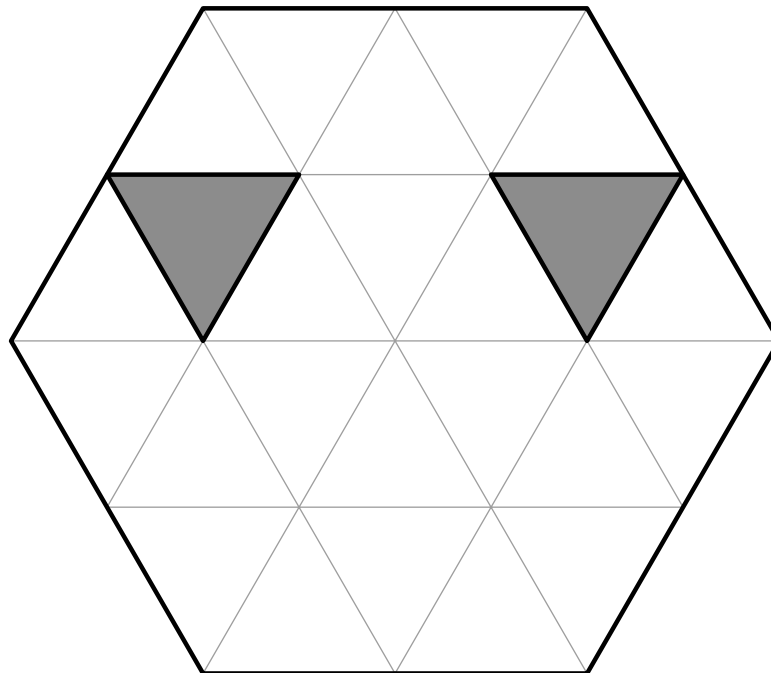
D



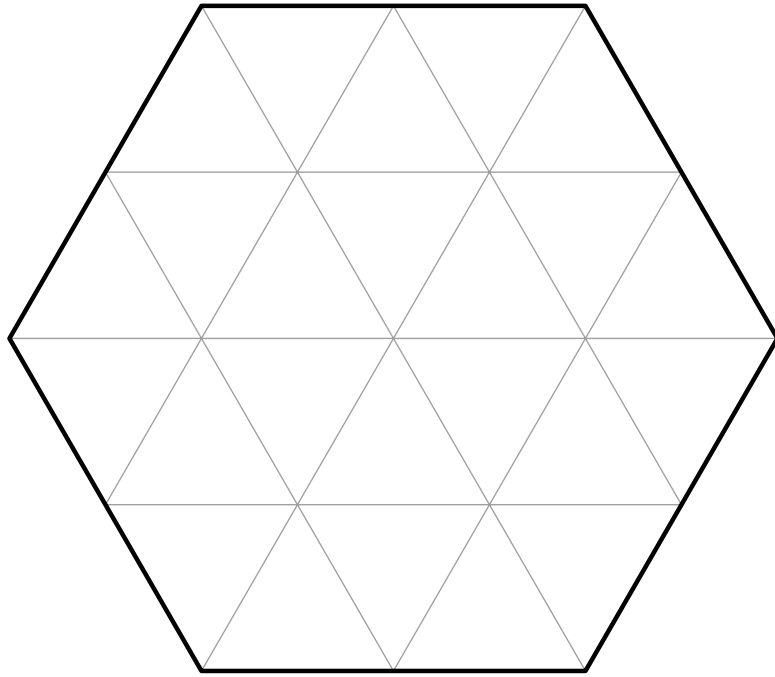
E



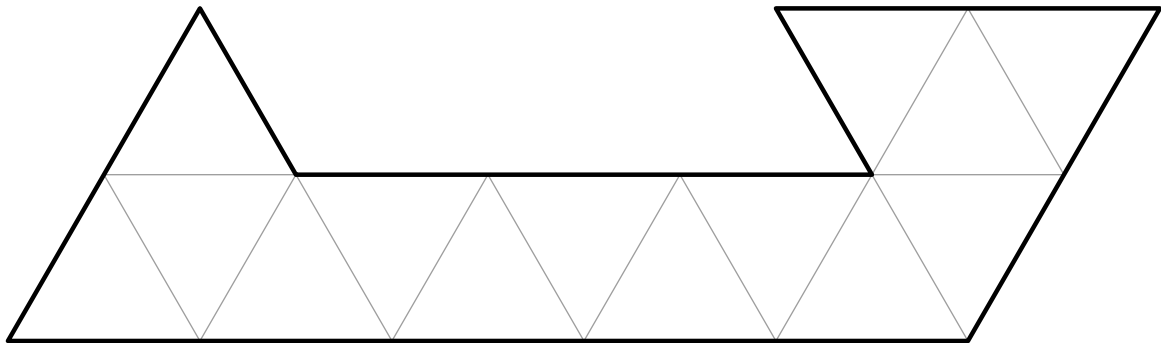
Problem 2: The gray triangles on boards F and G are holes, so they stay uncovered. Can the rest of board F be covered with blue pieces? Can the rest of board G?

F**G**

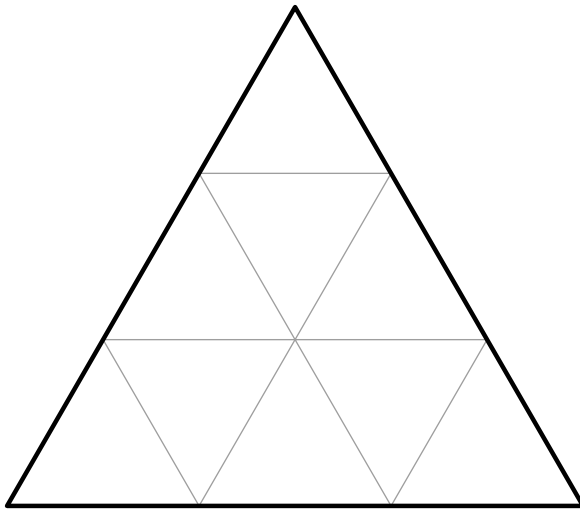
Problem 3: Mark two holes on board H by putting a green triangle on each. For which pairs of holes can the rest of board H be covered with blue pieces?

H

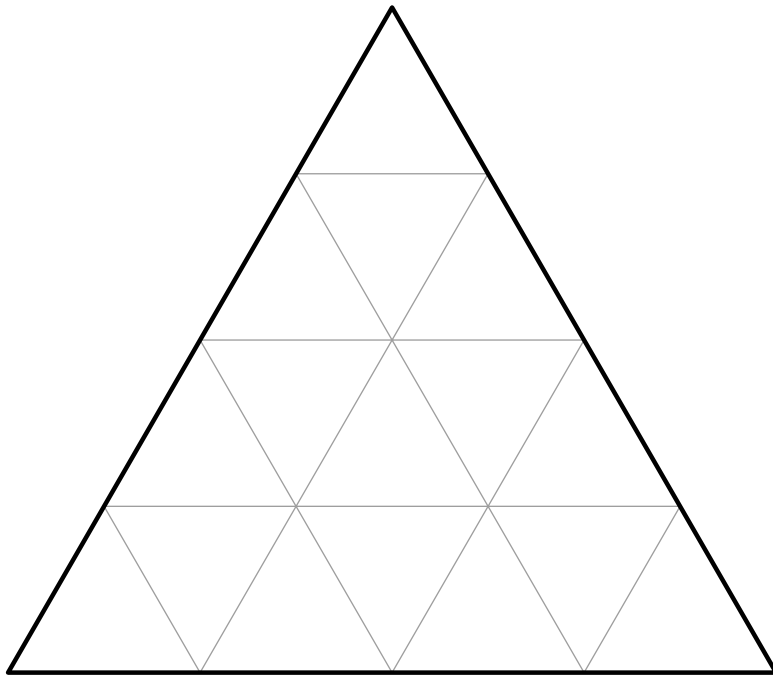
Problem 4: Can board J be covered with blue pieces?

J

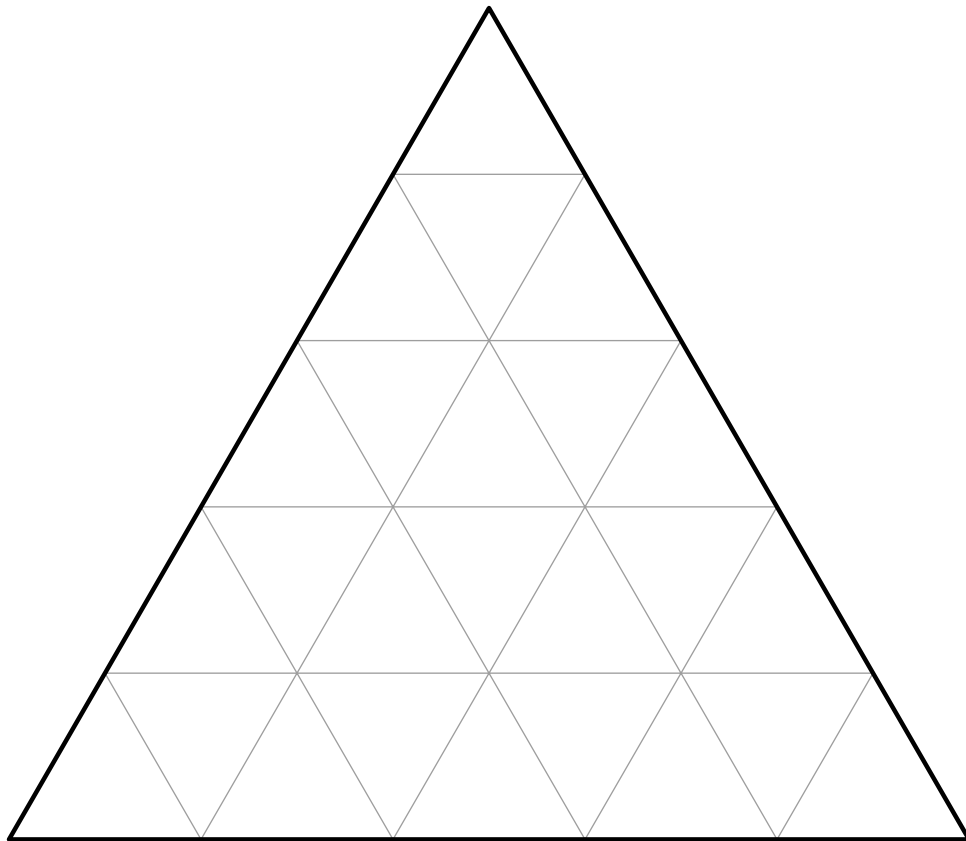
Problem 5: Cover each big triangle on pages 5 to 7 with blue pieces and green triangles. Use as few green triangles as you can.



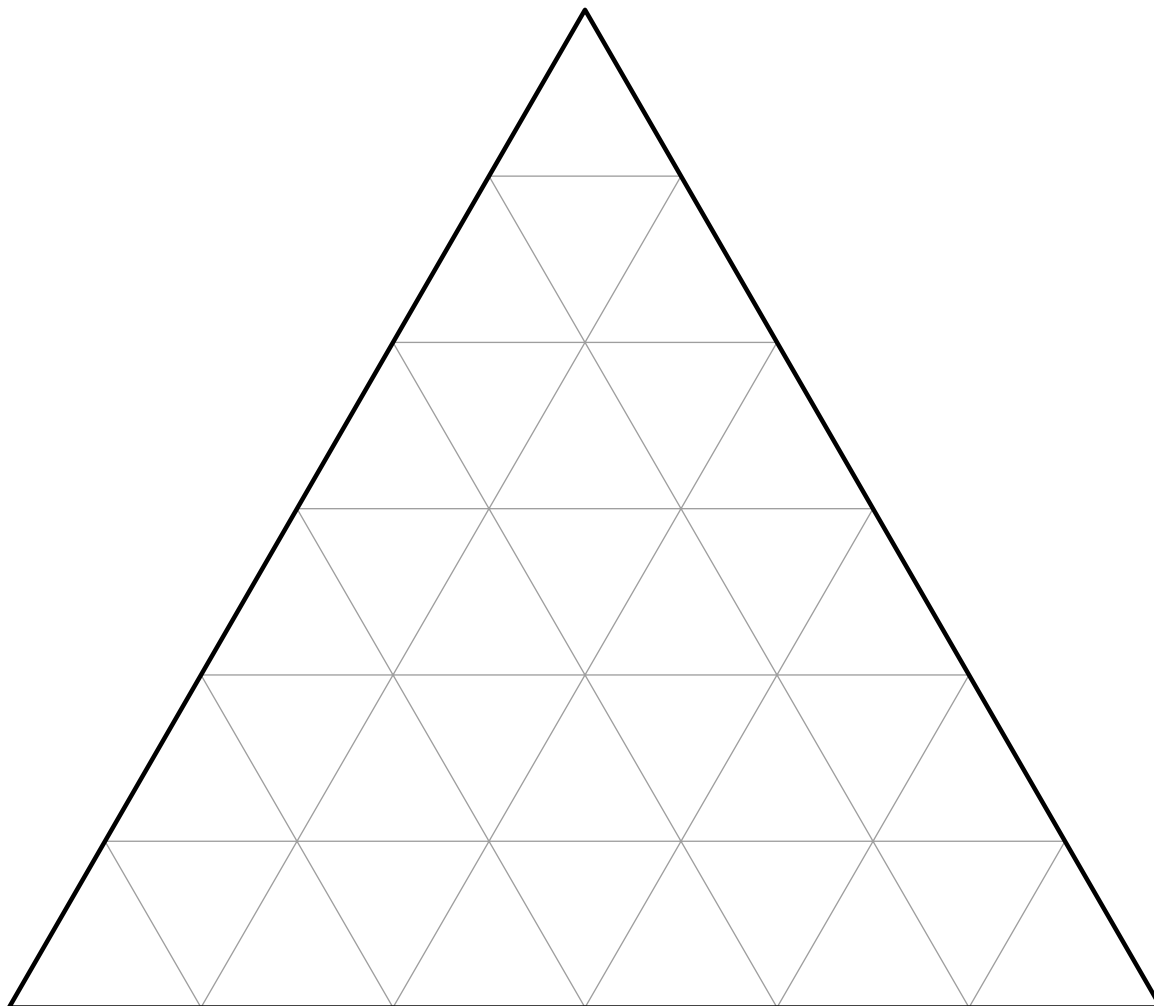
Green triangles used



Green triangles used



Green triangles used



Green triangles used

Problem 6: A big triangle has sides that are 10 small edges long. What is the smallest number of green triangles that can be used, together with blue pieces, to cover it? How do you know that nobody can use fewer? What if the sides are 100 small edges long?

Problem 7: Place as many purple pieces as you can on the big triangle with sides 4 small edges long. How many small triangles are left uncovered? Answer the same question for the big triangle with sides 5 small edges long. Is there any big triangle where purple pieces leave fewer triangles uncovered than blue pieces do?

Problem 8: Is there a board that purple pieces can cover but blue pieces cannot? Is there a board that blue pieces can cover but purple pieces cannot? You can draw boards on the grid below.

